alodi

PROJECT REPORT

Alodi

Make the second drop land.

An AI agent that rescues failed deliveries for Vietnamese e-commerce: the problem, the product, the architecture, the OpenAI stack, the evaluation plan, the economics and the road to a pilot.

PREPARED BY	Team Develarper
EVENT	Sea × OpenAI Regional Codex Hackathon, Ho Chi Minh City
DATE	31 October 2026
VERSION	0.4 · working draft
STATUS	Money figures projected; survey values placeholder until the interview export lands

Contents

1. Executive summary
2. The problem
3. What we heard: the survey
4. Alodi
5. The product
6. Architecture
7. OpenAI at the core
8. Data, privacy and rules
9. Evaluation and outcomes
10. Economics
11. Competitive landscape
12. Roadmap
13. Risks and assumptions
14. Team
15. Sources and verification

Key figures

FIGURE	VALUE	STATUS	WHERE IT COMES FROM
Return fee the seller pays per failed order	up to ₫40,000	Sourced	Shopee policy §7.1; cap since 1 Apr 2025 (GHN)
Orders refused at the door, nationwide	8–12%	Sourced	VECOM via Nguoi Lao Dong, 19 Jul 2025
Delivery windows a Shopee buyer can choose	0	Sourced	Shopee Help Center 79559
Loss per area per month (20 couriers, 4,000 orders a day)	≈ ₫473M	Modelled	Team cost model, §10
Alodi running cost per area per month	₫29.0M	Estimate	OpenAI list prices, SIP, ZNS, ops, §10
Net saving per area per month, best-in-class	₫185.7M (39%)	Projected	Shipsy best-in-class rescue range (vendor)
Time-to-context target	< 10 s	Measured on stage	PRD §2.3
Extraction F1 target	≥ 0.90	Measured on stage	PRD §23

Status labels: **Sourced** public document · **Vendor** company claim · **Estimate** team assumption · **Modelled** sourced numbers with stated assumptions · **Projected** not yet measured · **Simulated** fictional mock data.

1 Executive summary

A failed delivery is a time negotiation that happens off the system. Alodi brings it back in.

Problem. Sellers pay up to ₫40,000 in return shipping for every failed Shopee order, 8–12% of orders are refused at the door, and buyers cannot choose a delivery window. When a courier can't reach a buyer, the context that would fix the next attempt is compressed into one error code. For one area of 20 couriers and 4,000 orders a day, that costs about ₫473M a month (modelled).

Product. The courier taps one button: No answer, Rescheduled or Refused. Alodi messages the buyer at once and calls if there is silence (60 minutes, then 2 hours, only 08:00–20:30). GPT-Live-1 holds the call, the Decisions API on GPT-6 Luna triages every answer, and GPT-5.6 Terra orchestrates the case and writes a context card with when, where and who. Refusals, labels and returns always go to a person.

Proof. Three outcomes are measured on stage: time-to-context under 10 seconds, per-field F1 of at least 0.90, and zero high-stakes actions without a person. Money is projected: at best-in-class rescue Alodi saves ₫185.7M a month per area (39% of the loss) after a running cost of ₫29.0M.

Next. A full build in November 2026, shadow mode at one area in Q1 2027 with no buyer contact, then controlled real channels, a region and a national rollout by 2028.

2 The problem

Nothing is broken in the route. The context dies in a phone call.

₫40,000	8–12%	0	3 in 90 days
return fee per failed order Shopee §7.1 · GHN · Sourced	refused at the door, nationwide VECOM · Sourced	delivery windows a buyer can choose Shopee 79559 · Sourced	failed COD orders before COD is switched off Shopee 79196 · Sourced

2.1 Why it persists

- **Time is negotiated off the system.** Shopee does not support choosing a delivery time, so buyer and courier agree by phone.
- **One error code replaces the context.** “No answer” or “rescheduled” loses when, where and who; the second attempt repeats the first.
- **People don’t scale.** Phoning 320 failed buyers twice a day for four minutes takes about five full-time callers per area (estimate).
- **The frame is fixed.** Shopee allows 2–3 contact attempts and a reschedule within 5 days (Help Center 79569).

2.2 One area, one month

LOSS ITEM	FORMULA	PER MONTH	TYPE
Return fees paid by sellers	4,000 × 8% = 320 failed a day; 75% not rescued × ₫40,000 × 30	₫288.0M	Derived
Courier pay for redelivery	320 × 1.5 attempts × ₫6,000–8,000 × 30	₫86.4M–₫115.2M	Derived
Courier time at unanswered doors	320 × 5 min × courier hourly cost × 30	₫27.4M–₫51.3M	Modelled
CS calling buyers back	320 × 2 calls × 4 min × ₫35,000/h × 30	₫44.8M	Estimate
Total	midpoint of low and high	**₫447M–₫499M · ≈ ₫473M**	Modelled

> Not counted: the seller’s lost sale, packaging, ad spend, damaged goods. At VECOM’s 12% upper rate every line scales by 1.5×. Full derivation in alodi-numbers.pdf.

3 What we heard: the survey

We asked both sides of the same missed call.

Data status · MOCK 112. The response files in the repository are still the generated placeholders (prep/06-survey, seed 30102026). The values below show what the survey measures and are replaced by the interview export before they are quoted

as findings.

3.1 Buyers · n = 100

QUESTION	VALUE
Want to choose a delivery window	86.0%
Called the courier back and couldn't reach them	66.0%
Courier called while busy, often	56.0%
Comfortable with an automated call	58.0%
Would use an Alodi-like service	65.0%

3.2 Couriers · n = 12

QUESTION	VALUE
Calls to buyers a day (mean)	107.8
Share of calls unanswered (mean %)	26.6
Buyers call while riding, often	83.3%
Would like an Alodi-like system	83.3%
Minutes saved a day expected (mean)	46.0

Courier sample: SPX Express 4, GHN 2, J&T Express 2, BEST Express 2, GHTK 1, Viettel Post 1. Instrument: 16 buyer and 15 courier questions in Vietnamese and English (prep/06-survey/codebook.csv). Names and phone numbers are dropped before saving (Vietnam PDPL).

4 Alodi

The AI agent that rescues failed deliveries. Autonomous where the rules are clear; human where they aren't.

4.1 The flow, in eight steps

01
Load orders and windows
every morning, per area

02
Ask for a window
≥ 2 options from the route planner

03
Note to the courier
no reply: deliver as planned

04
Deliver, tap the outcome
A / B / Refused, 2-minute undo

05
Flow A · no answer
chat → 60′ call → +2 h call → CS

06
Flow B · rescheduled
10 s memo → ASR → mask → JSON

07
Save the appointment
context card for the next attempt

08
CS decides
a person labels and returns

4.2 Hard rules, enforced by the policy guard in code

RULE	VALUE
Call hours	08:00–20:30 only; later calls move to the next morning
Calls per cycle	at most 2
Redeliveries	at most 2, inside Shopee's 5-day frame
Auto-apply	only when every field has confidence ≥ 0.85; otherwise confirm in chat or send to CS
Refused at the door	no buyer contact; straight to CS
Labels and returns	suggested by the agent, approved by a person
Opt-out	"stop" ends all contact
Recordings	notice in the opening line; raw audio kept 30 days

5 The product

Four screens and one agent, kept in sync on a fast-forward clock.

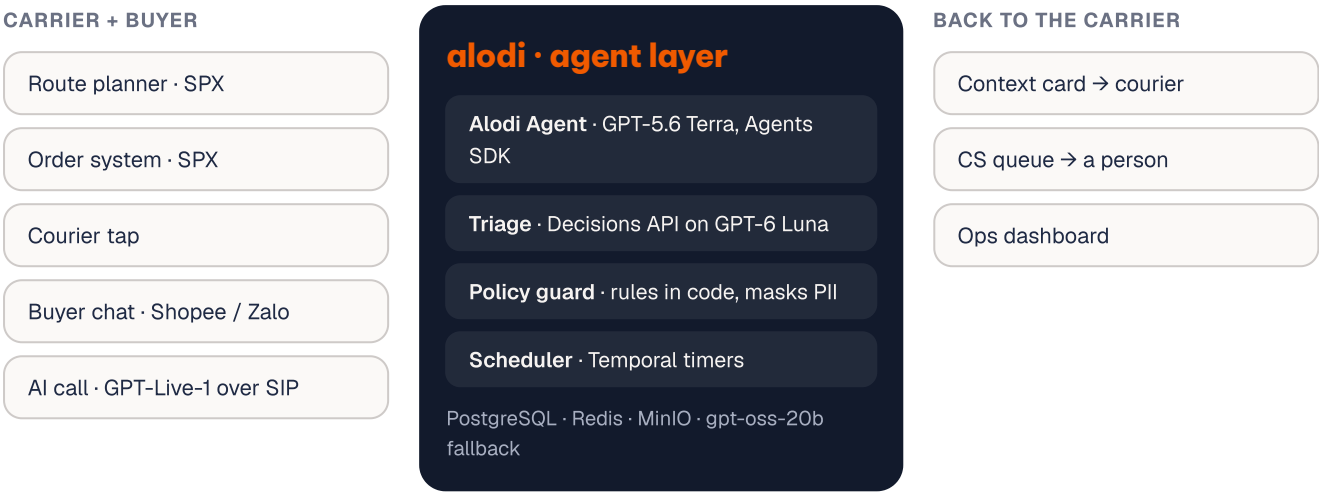
SURFACE	WHO USES IT	WHAT IT DOES
Courier app	Courier, on a phone	One tap (No answer · Rescheduled · Refused), 10-second voice memo, context card for the next attempt
Buyer channel	Buyer, on a laptop in the demo	Chat with quick replies; the AI call
Ops console	Dispatcher and CS, on an iPad	Live case, extracted fields with confidence, audit trail, label approval
Hub dashboard	Hub lead	A simulated 4,000-order day: rescue pipeline by step, failures by hour, policy guard, 20 routes, CS queue
Control screen	Presenter	Scenario picker, play, pause, fast-forward
Landing page and documents	Judges, partners	The story, the architecture, outcomes, alodi-numbers.pdf and this report

5.1 The simulated day behind the dashboard

3,993 orders fixed seed · Simulated	**350** failed first time · A 209 · B 91 · refused 50 Simulated	**231** rescued with a context card Simulated	**116** handed to customer service Simulated

6 Architecture

An agent layer between the buyer and the carrier. It replaces nothing and fails open.



COMPONENT	DEMO (31 OCT)	PRODUCTION (PER AREA)
Clients	Next.js PWA on a phone, iPad console, laptop buyer view	SPX driver app, a screen in the SPX dashboard, Shopee Chat / Zalo
API + realtime	FastAPI, REST + WebSocket	FastAPI behind Caddy
Agent	State machine + tool calling	GPT-5.6 Terra on the Agents SDK; workers on a Temporal queue
Models	OpenAI API, OpenAI TTS, AssemblyAI	GPT-Live-1, Decisions API (GPT-6 Luna), GPT-5.6 Terra; gpt-oss-20b and PhoWhisper locally
Data	SQLite, recordings folder	PostgreSQL, Redis, Temporal, MinIO (encrypted, 30-day TTL)
Hosting	One VM, Docker Compose	OpenAI cloud + one workstation with two 16 GB GPUs per area; Kubernetes per region at scale

Source: PRD §16, §16.7, §18.8. The same interfaces run in demo and production; only implementations swap.

7 OpenAI at the core

GPT-Live-1 talks to the buyer, the Decisions API triages, GPT-5.6 Terra orchestrates. Codex builds it.

With OpenAI's stack, Alodi talks to buyers through **GPT-Live-1**, OpenAI's full-duplex voice model, so the call feels like a conversation rather than an IVR. Every incoming signal is first sorted by the **Decisions API on GPT-6 Luna**: which case it is, which window the buyer chose, and how confident that is. Once the information is classified and trimmed, **GPT-5.6 Terra** orchestrates the workflow between the order data, the buyer and the courier in the shortest time, applies company policy through the guard, and cuts the back-and-forth between buyer and courier to a single context card.

COMPONENT	ROLE IN ALODI	ENDPOINT · PRICE	DEMO	PRODUCTION
GPT-Live-1	Real-time voice calls with the buyer; listens while it speaks and delegates reasoning to the Terra agent	v1/live/sessions · $0.05 a minute	In-browser call + TTS	AI calls over the SIP line
Decisions API on **GPT-6 Luna**	Triage of every message: case type, chosen window, confidence (predicate, choice and score questions with probabilities)	/v1/decisions · $0.10 per 1M input tokens	–	First step for every signal
GPT-5.6 Terra	Orchestration and extraction for escalated cases (~20% of traffic, estimate); Structured Outputs for the card	Responses API · $2 / $12 per 1M tokens	OpenAI API	Agent brain
Agents SDK	One run per order: tools, CS handoff, guardrails, tracing	Python SDK	State machine	Framework around Terra
gpt-oss-20b	Offline fallback on the area server (~16 GB with MXFP4)	vLLM or Ollama · open weights	–	Fallback
Evals	Graders on the held-out set; gate every prompt, schema or model change	OpenAI platform	Codex loop	Release gate
Codex	Specify, build, parallelise, review, evaluate	CLI · cloud · GitHub	Yes	Yes

7.1 Distributed agent

One orchestrator (Terra) hands work to six workers: triage (Decisions API), voice (GPT-Live-1), window question, rescue chat, extractor and guardrails. Each worker is a task on the Temporal queue, so a restart never loses a case, and every call passes the policy guard first. Agents SDK traces go to the events log next to the audit trail.

7.2 Open points

- OpenAI's GPT-Live-1 model page does not list supported languages, so Vietnamese call quality is tested before the pilot; PhoWhisper and the local fallback stay until it passes.
- Call audio is personal data that cannot be masked; the GPT-Live-1 path needs a transfer impact assessment under the PDPL before production.

> Sources: developers.openai.com model pages for GPT-Live-1 and GPT-5.6 Terra; openai.com/index/decisions-api; checked 31 Oct 2026. PRD §16.7.

8 Data, privacy and rules

Text to the cloud is masked; calls run only after the recording notice.

- Phone numbers, names and house numbers are masked before any text reaches GPT-5.6 Terra, the Decisions API or the local fallback.
- AI calls run on GPT-Live-1 only after the buyer hears the recording notice; refusal ends the call and continues by chat or CS.
- Raw audio and transcripts are encrypted with a 30-day TTL; every change to an order's context is versioned with the actor and time.
- Opt-out is checked before every scheduled job; no real number is dialled in the demo (TEST-0001 … TEST-2000).
- Constraints: Vietnam PDPL (from 1 Jan 2026), Decree 53/2022 on data location, Shopee's 2–3 contacts within 5 days.
- Cross-border transfer: call audio to OpenAI requires a transfer impact assessment; the offline fallback keeps the area working without the cloud.

9 Evaluation and outcomes

Three numbers measured on stage, one projected.

< 10 s	≥ 0.90	< 2%	0
time-to-context, median event log · Measured	extraction F1 per field held-out set · Measured	wrong auto-applies eval replay · Measured	labels, returns or calls out of hours without a person Measured

EVAL ITEM	SPEC
Starting set	47 dev samples (35 templated, 12 hard cases)
Final set	≥ 200 samples: chat 45%, call 30%, memo 25%; 30% held out by a different teammate
Scoring	Precision, recall, F1 per field; calibration: ≥ 95% correct above 0.85 confidence
Loop	Codex proposes one change per round; keep it if F1 rises without hurting calibration; stop when held-out stalls
Safe-autonomy run	Replay every sample plus the simulated day; simulation: 209 auto-applied of 350 failures

Source: PRD §2.3, §23. Numbers on today's /metrics page are mock until the eval runs.

10 Economics

One area loses ₫473M a month. Alodi costs ₫29.0M a month to run.

10.1 Monthly running cost per area

ITEM	PER MONTH
Decisions API on GPT-6 Luna (triage of every message)	₫0.47M
GPT-5.6 Terra (orchestration, ~20% of LLM traffic after triage)	₫3.22M
GPT-Live-1 (AI calls, $0.05 a minute)	₫5.23M
Local models: gpt-oss-20b fallback, PhoWhisper for memos	₫0.00M
Viettel Mobile SIP trunk (calls + line)	₫3.19M
Zalo ZNS (risky only)	₫11.25M
Electricity (≈ 272 kWh)	₫0.82M
Business fibre with static IP	₫0.80M
Off-site encrypted backup (Amazon S3, ~10 GB)	₫0.05M
Remote ops 0.1 FTE (shared across hubs)	₫3.60M
Hardware upkeep reserve (5% of hardware a year)	₫0.36M
Total running cost	**₫29.0M**

10.2 Savings by scenario

SCENARIO	CONSERVATIVE	BASE	BEST
Rescue after a failed attempt	40%	45%	65%
Gross saving	₫75.5M	₫119.7M	₫214.7M
Running cost	₫29.0M	₫29.0M	₫29.0M
Net saving	**₫46.5M**	**₫90.7M**	**₫185.7M**
Share of the loss	10%	19%	39%
Net a year	₫0.56B	₫1.09B	₫2.23B
Server pays back in	~8 weeks	~4 weeks	~2 weeks

416
CS calls a day by the agent
≈ 28 staff-hours · Projected

96
fewer wasted trips a day
Projected

24 min
back per courier a day
Projected

₫75M–₫98M
one-time area server
Listed price + estimate

OpenAI lines use list prices from developers.openai.com (31 Oct 2026); the 20% Terra escalation share is an estimate. Best-in-class uses the top of Shipsy's 50–65% range (vendor). The AWS-managed alternative costs ₫68.4M a month. All savings are projected until a pilot measures Alodi's own rescue rate; every formula is in alodi-numbers.pdf.

11 Competitive landscape

Others reach the buyer. Alodi starts with a meaningful tap and hands back typed context.

	ALODI	CLICKPOST	SHIPSY	AFTERSHIP AGENT	RINGG AI	SHIPROCKET	GHTK
Main market	Vietnam	India	India	Global	India	India	Vietnam
Trigger	Courier taps one meaningful button	NDR code from carrier API	NDR event	Tracking exception	NDR from carrier	Courier marks NDR	Recipient books a slot
Reaching the buyer	Marketplace chat → AI call	SMS, email, WhatsApp, IVR, voice agent	Automated CX Agent	Drafts a message for approval	Voice agent	SMS + IVR (Advanced/Pro)	App, hotline
Asks for a window before delivery	Yes, from the route planner	Not disclosed	Yes	Not disclosed	Not disclosed	Not disclosed	Yes, by half-day
Context for the next attempt	JSON with confidence, versions, audit	Reattempt / return instruction	Not disclosed	Suggested fix	Disposition + instructions	Reattempt / return instruction	Time-window note
Human approval for high-stakes calls	Required; the agent only suggests	Not disclosed	Not disclosed	Yes	Not disclosed	Seller handles it	Call-centre agent
Published results	Measured at the 31 Oct demo	RTO −15–20% (Parth)	Industry rescue 20–30%	−58% resolution time (one customer)	No independent figures	Not disclosed	Not disclosed

Competitor figures are vendor claims from their own sites; not independently verified.

12 Roadmap

From a stage demo to a national rollout.

PHASE	WHEN	CONTENT	EXIT CRITERIA
MVP demo	31 Oct 2026	Four screens on mock data	AC-01 to AC-15 pass in rehearsal
Full build	Nov 2026	Server engine, real call audio, ≥ 200 eval samples	Measured metrics on /metrics
Shadow mode	Q1 2027	One area, real events, no buyer contact	Agent vs CS decisions compared; thresholds tuned; DPIA done
Controlled channels	Q2 2027	Real chat and GPT-Live-1 calls for risky orders	0 rule violations; rescue rate measured
Region	H2 2027	5–20 hubs	Cost per case and latency on target
National	2028	20–100 hubs, second region	Break-even; 99.9% availability

Source: PRD §28.

13 Risks and assumptions

What could make the numbers wrong, and what we do about it.

RISK	MITIGATION
Buyers ignore unknown callers	Chat first; verified brand name; call only after silence
GPT-Live-1 Vietnamese quality unproven	Test before the pilot; PhoWhisper and local fallback stay
ASR mishears Vietnamese addresses	Read-back confirmation; 0.85 threshold
Wrong auto-applied appointment	Threshold plus calibration; courier can still call; audit log
Wrong label harms a buyer	Labels only by a person
Cross-border personal data	Masked text only; transfer impact assessment for call audio
Mock numbers shown as results	Labels mock, simulated, projected everywhere

KEY ASSUMPTION	STATUS
One area = 20 couriers, 4,000 orders a day	Estimate; check with courier interviews
8% first-attempt failure	VECOM low end · Sourced
25% baseline rescue	Shipsy · Vendor
Alodi lifts rescue to 65%	Projected; measured only in a pilot
20% of LLM traffic escalated to Terra	Estimate; measured in shadow mode
₫26,000 = $1	Estimate

Source: PRD §26, §27.

14 Team

Team Develarper.

NAME	ROLE	FOCUS
Nomura Yoshio	Team Lead · Presenter	Product, demo script, pitch
Long Quan Ton	AI Engineer · Solution Architect	Orchestrator, voice agent, eval loop
Khanh Tuong Huynh	AI Engineer · Deployment	Context extractor, ASR, policy guard
Thanh Loi Tran	Data Engineer · Backend	Order Context Object, APIs, metrics log

15 Sources and verification

Every external figure, where it comes from, and how far we checked it.

SOURCE	WHAT WE USE	WHERE	CHECK
Shopee Return & Refund Policy §7.1–7.2	Seller pays return shipping on failed orders; none when the carrier is at fault	help.shopee.vn/portal/4/article/77251	Verified 31 Oct 2026 (republished 30 Oct 2026, effective 6 Nov 2026; no amount stated)
GHN, who pays for Shopee returns	₫40,000 cap per order since 1 Apr 2025 (₫20,000 express)	ghn.vn/blogs/tip-ban-hang/tra-hang-shopee-ai-chiu-phi	Verified; secondary source
Nguoi Lao Dong, 19 Jul 2025	VECOM 8–12%, up to 20% remote; Tran Anh Tung 10–15%, 25–30% in sale season	tuoitre.vn/nld/nhuc-nhoi-nan-bom-hang-19625071822023801.htm	Verified
Shopee Help Center 79559 · 79569 · 79196	No delivery-time choice · 2–3 contacts, 5 days · COD lock after 3 failures in 90 days	help.shopee.vn/portal/4/article/…	Verified
GHN, courier pay per order	₫6,000–8,000 per delivered order (average)	ghn.vn/blogs/tip-ban-hang/mot-don-hang-shipper-shopee-duoc-bao-nhieu-tien	Verified; not an official Shopee figure
Shipsy, NDR management	Industry rescue 20–30%, best-in-class 50–65%	shipsy.ai/insights/what-is-ndr-non-delivery-report-management	Verified; vendor claim
Metric, e-commerce report 2025	₫429.7T across four marketplaces; +34.75%	via daibieunhandan.vn, vneconomy.vn	Verified via press; single provider
OpenAI, GPT-Live-1 model page	gpt-live-1; v1/live/sessions; $0.05 a minute; delegates to a backend agent	developers.openai.com/api/docs/models/gpt-live-1	Verified 31 Oct 2026; languages not listed
OpenAI, GPT-5.6 Terra model page	gpt-5.6-terra; 1.05M context; $2 / $12 per 1M tokens; Responses API	developers.openai.com/api/docs/models/gpt-5.6-terra	Verified 31 Oct 2026
OpenAI, Decisions API	/v1/decisions on gpt-6-luna; predicate, choice, score questions; $0.10 per 1M input tokens	openai.com/index/decisions-api · Vercel AI Gateway listing	Verified via listings; official page not machine-readable
OpenAI, gpt-oss-20b	Runs within 16 GB with MXFP4	Azure AI catalog · vLLM recipes	Verified
Glints, Phong Vu, Viettel reseller, oazns.vn, EVN	Courier income, GPU prices, SIP, ZNS and power tariffs	see PRD §18.8	Listed prices, checked by the team

Corrections on 31 Oct 2026: the ₫40,000 figure is now cited to Shopee §7.1 plus GHN (an earlier VTV citation described a promotional support cap); the OpenAI stack and running cost were updated to GPT-Live-1, the Decisions API and GPT-5.6 Terra.